CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

Office: S9-14/Tel: 66017501

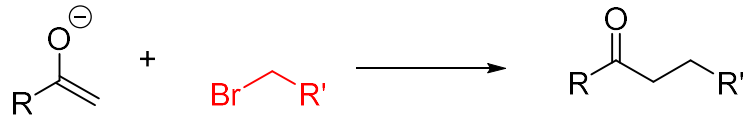
chmzhu@nus.edu.sg

Claisen condensations

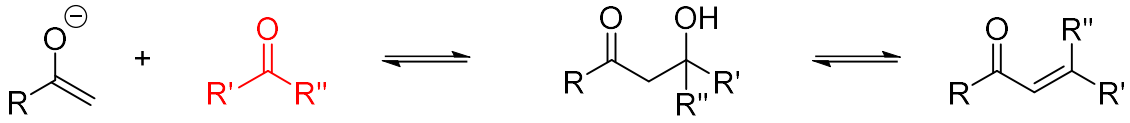
Readings: Solomons 13th ed, Chapter 19 and Clayden 2nd ed; Chapter 26

Reactions of enolates

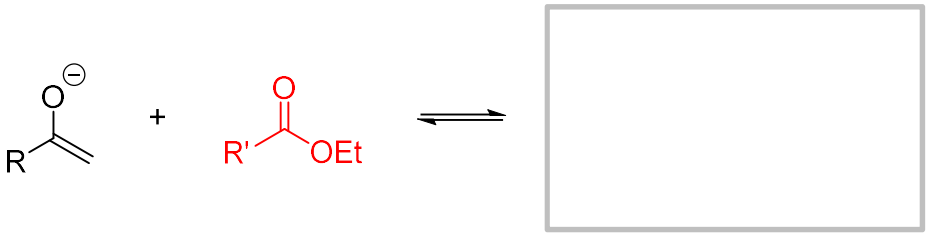
1. α-alkylation reaction



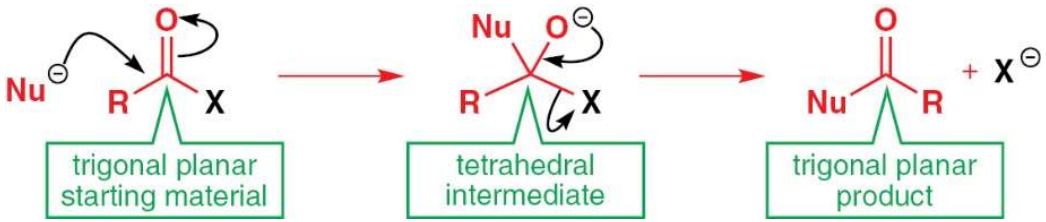
2. Aldol reactions: Enolate + aldehyde or ketone



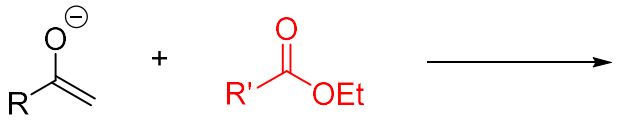
3. Claisen reaction: Enolate + ester



Nucleophilic acyl substitution reaction



Task 1 (general reaction): Claisen condensation = reaction between an enolate and an ester. Provide the product for the following reaction

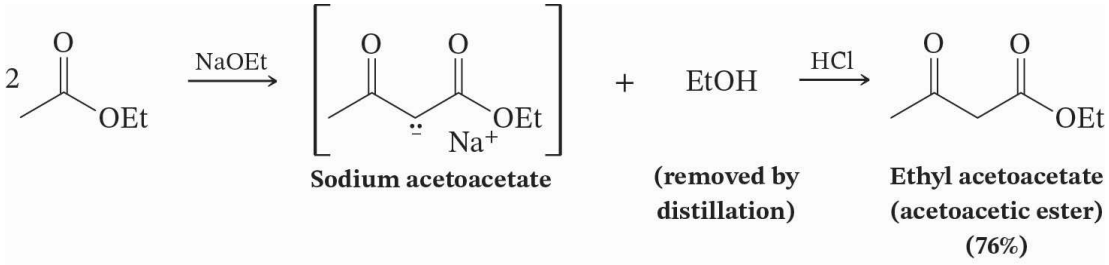


Nucleophilic acyl substitution reaction

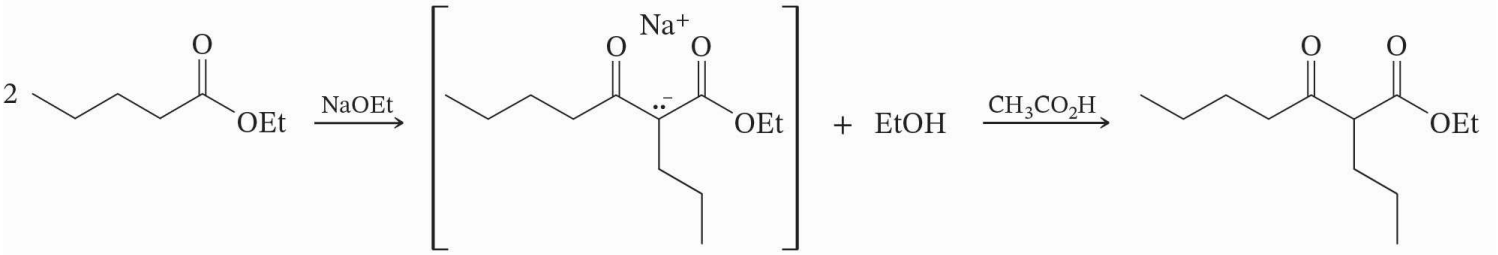
Task 2 (actual reaction): when ethyl acetate is treated with sodium ethoxide, what is the Claisen self-condensation product?



1. Claisen self-condensation

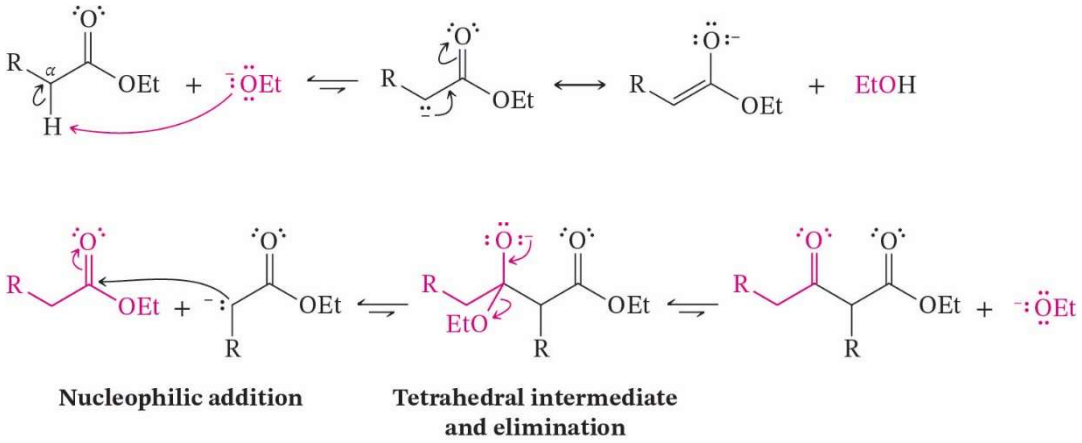


Sodium acetoacetate | **(removed by distillation)** | **Ethyl acetoacetate (acetoacetic ester) (76%)**



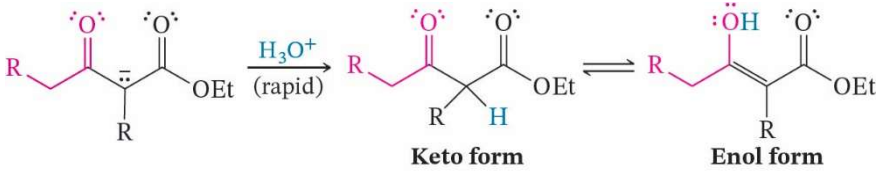
Ethyl pentanoate | **Ethyl 3-oxo-2-propylheptanoate (77%)**

Claisen self-condensation mechanism



Deprotonation of the product drives the reaction

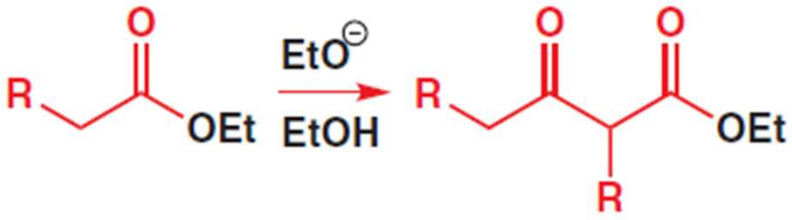
(1 equiv base is needed)



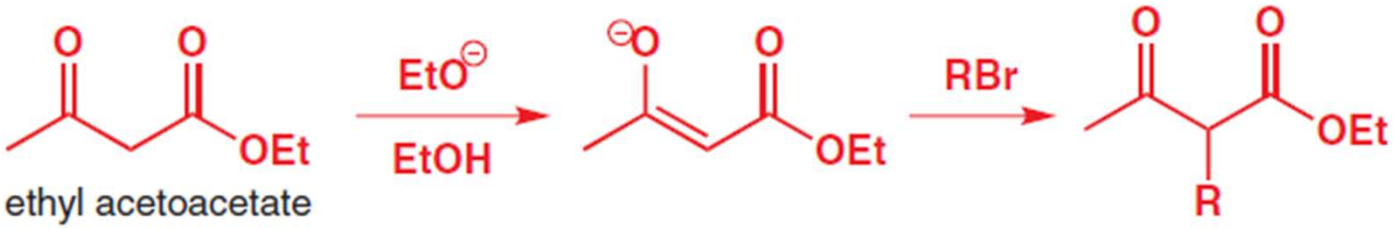
Reaction workup

Comparing the following two reactions:

Claisen self-condensation

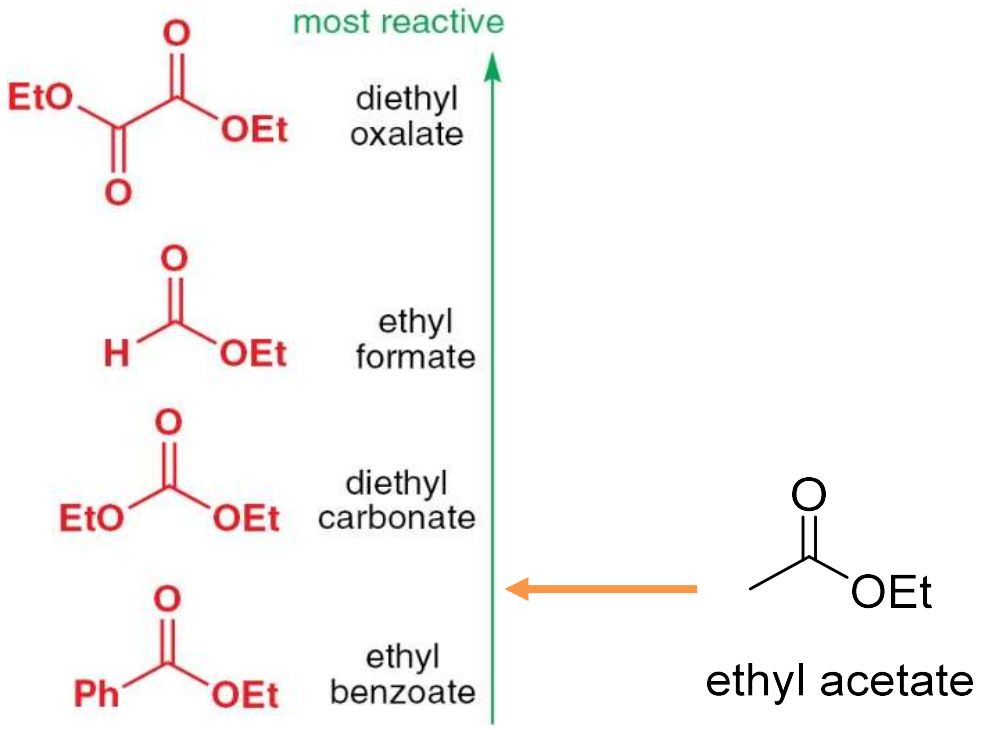


Alkylation of 1,3-dicarbonyl compounds



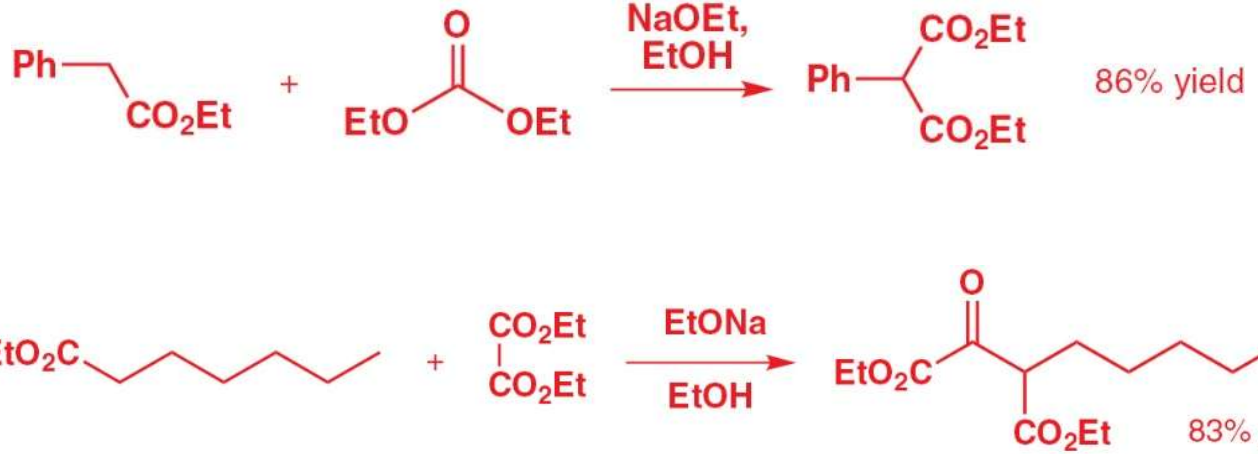
2. Crossed-Claisen condensations

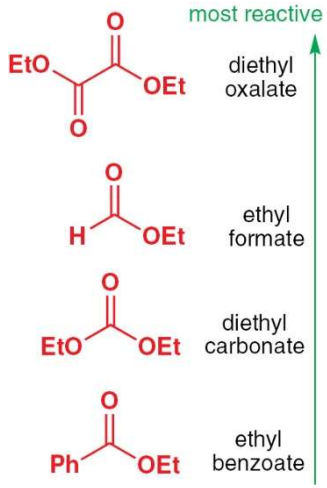
Esters that cannot enolize, ranked in order of reactivity (as electrophile)



Crossed-Claisen condensations

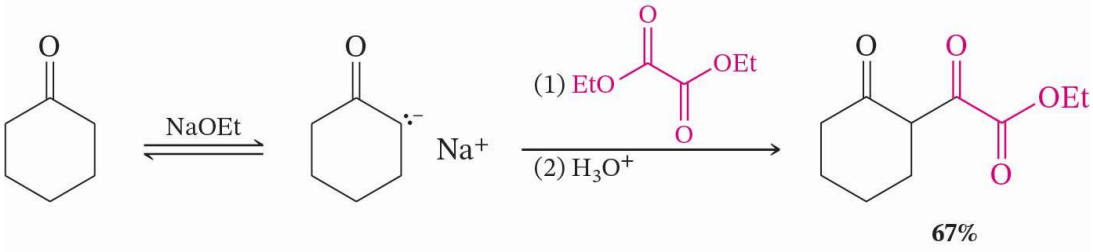
Ester + esters (or carbonate)

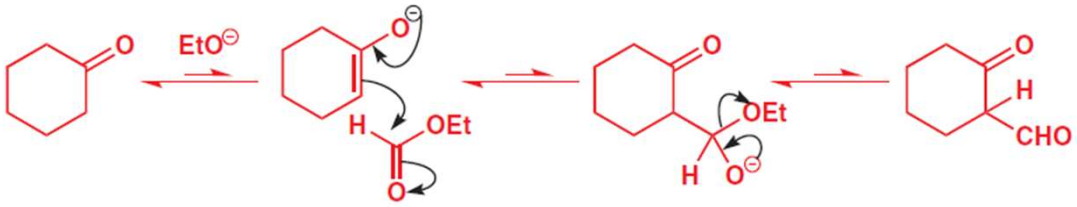


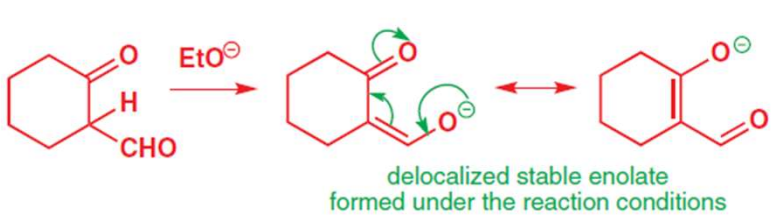


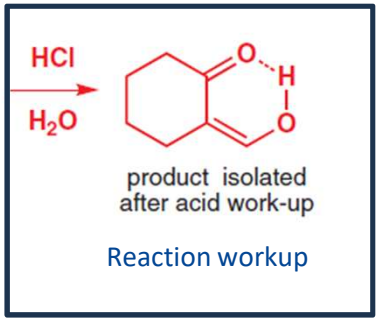
Crossed-Claisen condensations: examples

Ketone + esters (or carbonate)



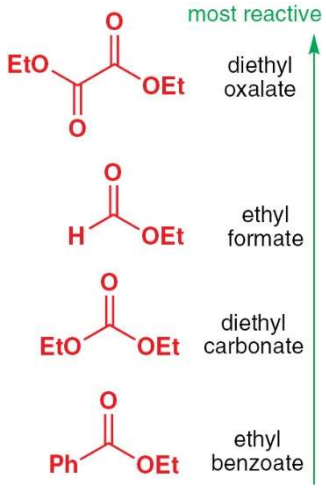




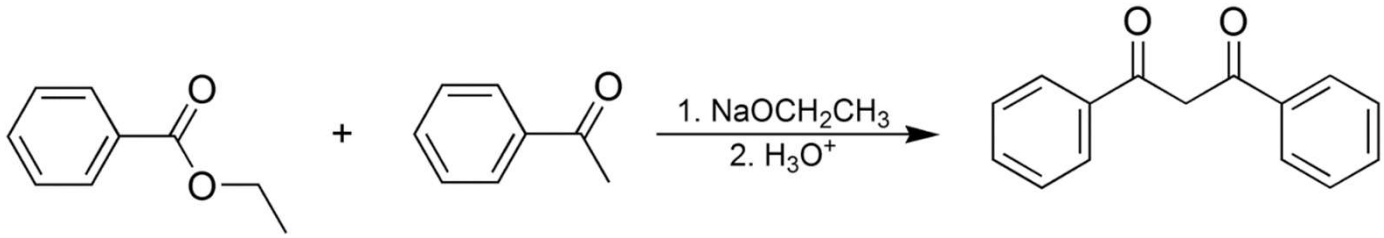


Deprotonation of the product drives the reaction

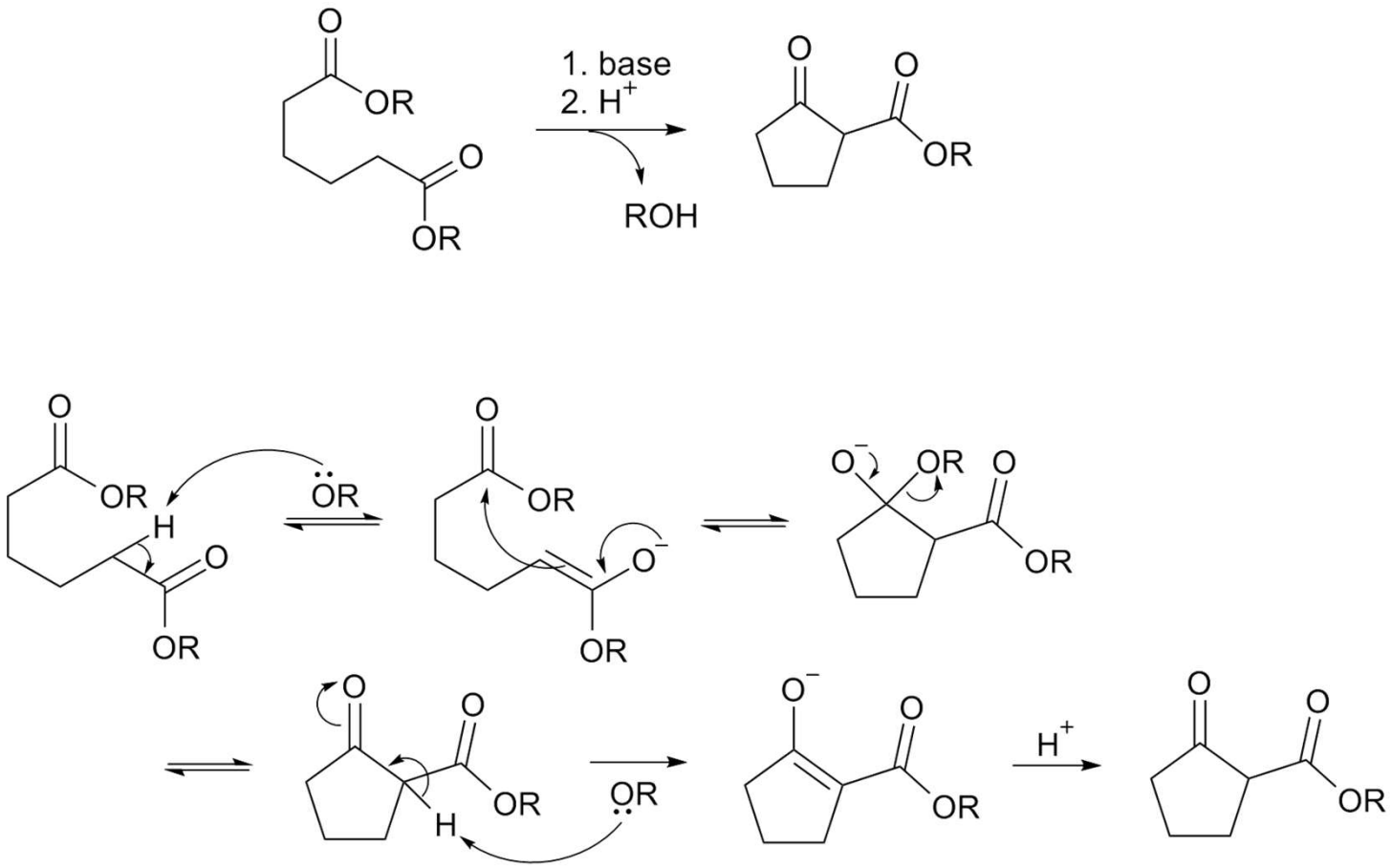
(1 equiv base is needed)



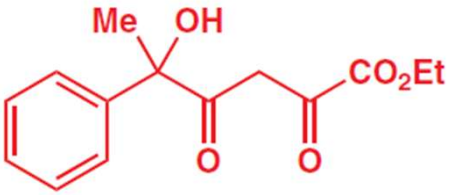
Practice: Explain why the following reaction affords the crossed Claisen product, not the Aldol product of the ketone.

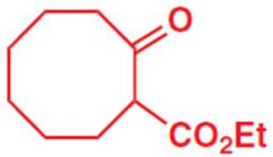


Intramolecular Claisen reaction (Dieckmann condensation)



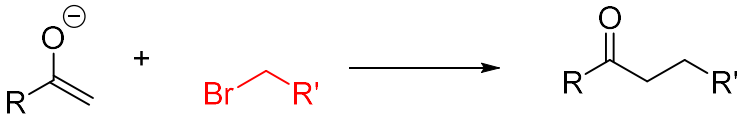
Practice: Proposed Claisen reactions to synthesis:



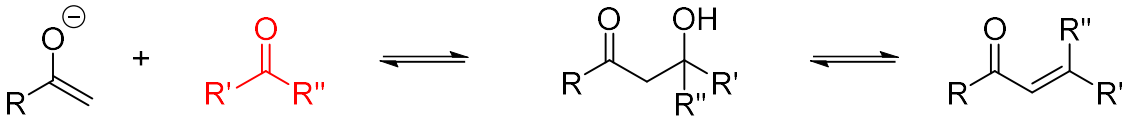


Reactions of enolates

1. α-alkylation reaction



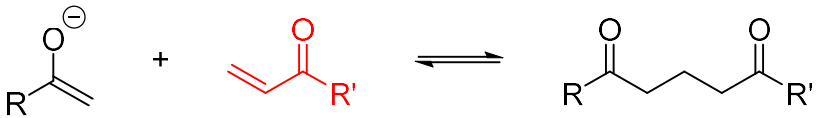
2. Aldol reactions: Enolate + aldehyde or ketone



3. Claisen reaction: Enolate + ester



4. Conjugate addition (Michael addition): Enolate + α,β-unsaturated carbonyl compounds



L5

Claisen condensation

Carbon–carbon bond forming reaction that occurs between one ester and one carbonyl compound in the presence of a strong base.

- The reaction produces a β-keto ester or a β-diketone.
- Self condensation of esters
- Cross condensation between one “activated” ester and another carbonyl compound
- Intramolecular reaction (Dieckmann condensation)